Machine Intelligence 1 Tutorial

01 Introduction and Learning Paradigms

Fachgebiet Neuronale Informationsverarbeitung (NI)

WiSe 25/26

Setting expectations

Learning Paradigms

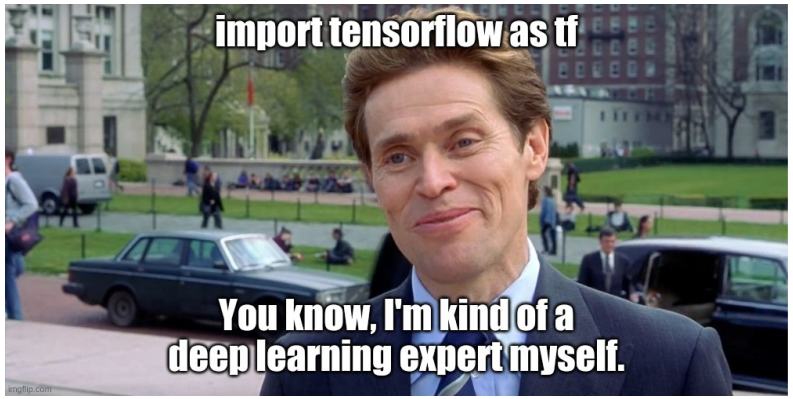
- Supervised learning
- Unsupervised learning
- Reinforcement learning
- Summarized overview

Connectionist Neuron (Perceptron)

- Components of the connectionist neuron
- Linear vs. non-linear transfer functions
- Shortcut notation for weights and bias

References

What not to expect from this course



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breadth of topics, not necessarily state of the art
theoretical aspects of the learning algorithms

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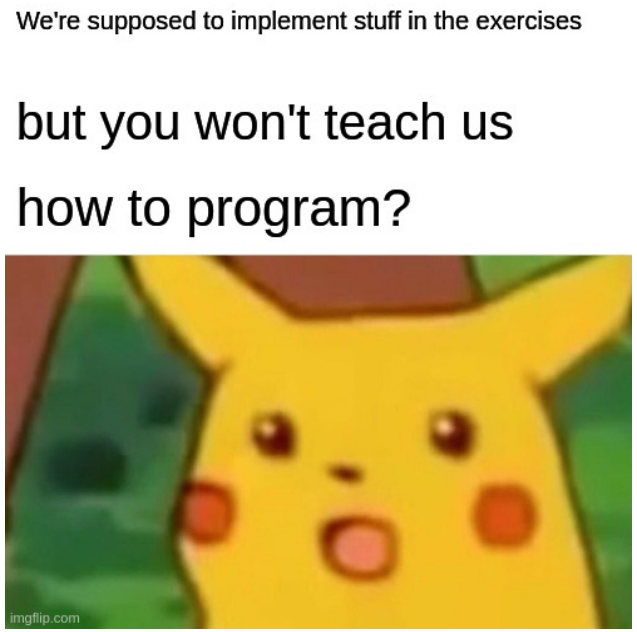

Learning Paradigms

- Supervised learning
- Unsupervised learning
- Reinforcement learning
- Evolutionary learning

Differences: Data & objective model.

Learning Paradigms: Supervised learning

example image recognition

Learning Paradigms: Supervised learning

We are given data that is made up of tuples.
Supervised methods fit a function that maps $\underline{\mathbf{x}}$ to some attribute $\underline{\mathbf{y}}$ (map observations to labels/ground truth).

example image recognition

Fit a function that maps image pixels to recognizing the object in the image from a set of possible objects (e.g. cat, dog, tiger, duck, frog, fish).

Supervised learning

essentially function fitting.
<u>Data</u>:

Consider this dataset of tuples:

$$\left(\underline{\mathbf{x}}^{(1)}, \underline{\mathbf{y}}_T^{(1)}\right), \ldots, \left(\underline{\mathbf{x}}^{(\alpha)}, \underline{\mathbf{y}}_T^{(\alpha)}\right), \ldots, \left(\underline{\mathbf{x}}^{(p)}, \underline{\mathbf{y}}_T^{(p)}\right)$$

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Objective model:

find a function that maps observations to their label(s)

- a discriminative function $\underline{\mathbf{y}} = \underline{\mathbf{f}}(\underline{\mathbf{x}})$,
  $\underline{\mathbf{f}} : \mathbb{R}^N \to \mathbb{R}^M$
- a conditional distribution $P(\,\underline{\mathbf{y}}\,|\,\underline{\mathbf{x}}\,)$.

Unsupervised learning

interesting directions and/or structure in the data using only observations $\underline{\mathbf{x}} \in \mathbb{R}^N$.

Data:

A dataset of observations:

$$\underline{\mathbf{X}} = \begin{pmatrix} | & | & & | & & | \\ \underline{\mathbf{x}}^{(1)} & \underline{\mathbf{x}}^{(2)} & \dots & \underline{\mathbf{x}}^{(\alpha)} & \dots & \underline{\mathbf{x}}^{(p)} \\ | & | & & | & & | \end{pmatrix} \in \mathbb{R}^{N \times p}$$

where

$p \mathrel{\widehat{=}}$ no. of observations (often i.i.d.)

$N \mathrel{\widehat{=}}$ no. of dimensions

Unsupervised learning

Objective model:

- the underlying distribution $P(\underline{\mathbf{x}})$ that generated this data (e.g. density estimation),
- $\underline{\mathbf{z}} := \underline{\mathbf{f}}(\underline{\mathbf{x}})$, where $\underline{\mathbf{z}}$ is a measure of
  - possible structure such as clustering or grouping in the data ($\underline{\mathbf{z}} \in 0, \ldots, K-1$),
    and/or
  - possible directions in the data, by finding another continuous space for describing this data.
    Example: dimensionality reduction, $\underline{\mathbf{z}} \in \mathbb{R}^M$ with $M < N$.

Reinforcement learning (RL)

RL is about

- what actions to take
- in which situations
- in order to maximize some reward.

Learning happens through an interaction of the agent with its environment.

Data in RL

Our data consists of a sequence. Each time step is described by

- a state $\underline{\mathbf{x}} \in \mathcal{X}$ or $\underline{\mathbf{x}} \in \mathbb{R}^N$,
  e.g. $\mathcal{X} := \{\underline{\mathbf{x}}_1, \ldots, \underline{\mathbf{x}}_S\} \subset \{0,1\}^S$ (1-out-S encoding)

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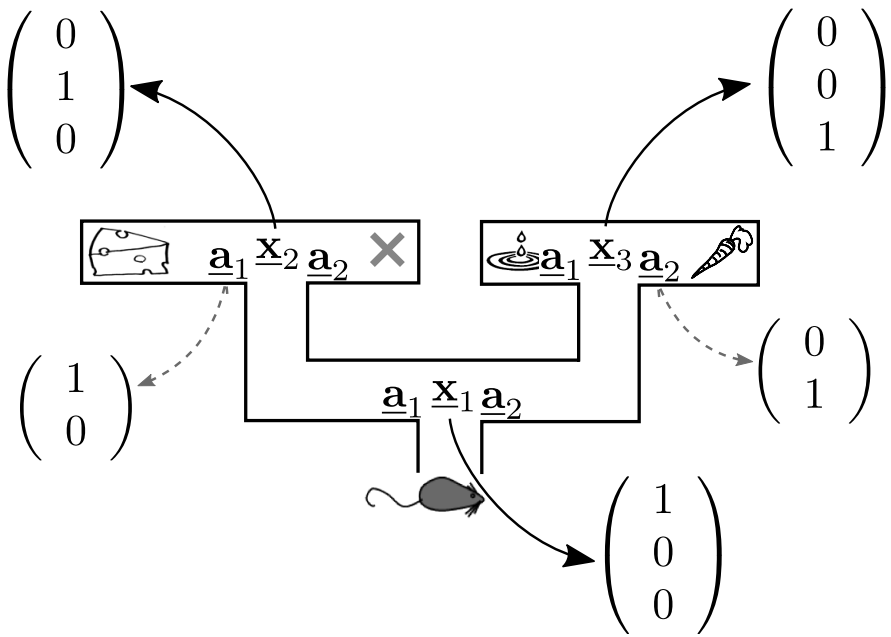
- a reward $r \in \mathbb{R}$ or $r \in \{0,1\}$, e.g. $r \in \{\text{“cheese”}, \text{“no cheese”}\}$.

$$\left\{\underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}, r^{(t)}\right\}_{t=0}^{p} = \left(\underline{\mathbf{x}}^{(0)}, \underline{\mathbf{a}}^{(0)}, r^{(0)}\right), \ldots, \left(\underline{\mathbf{x}}^{(p)}, \underline{\mathbf{a}}^{(p)}, r^{(p)}\right)$$

Q: *Can we still assume i.i.d.?*

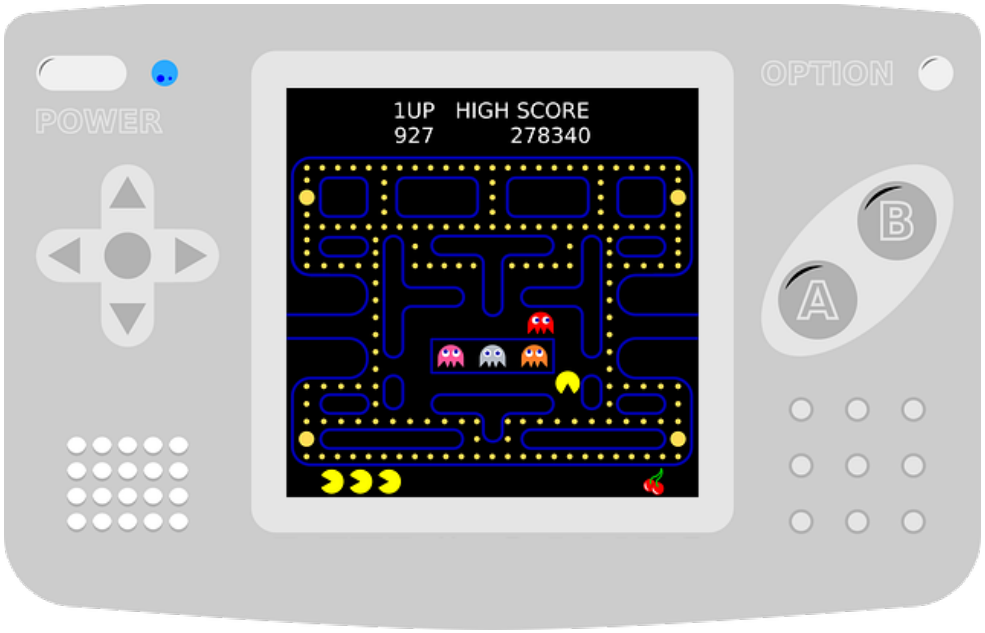
Example

- $\underline{\mathbf{x}} \mathrel{\widehat{=}}$ position in a maze
- $\underline{\mathbf{a}} \mathrel{\widehat{=}}$ movement direction (velocity), e.g. turn left/right
- $r \mathrel{\widehat{=}}$ found the cheese, found smaller piece of cheese. A negative reward, "punishment" is also possible.



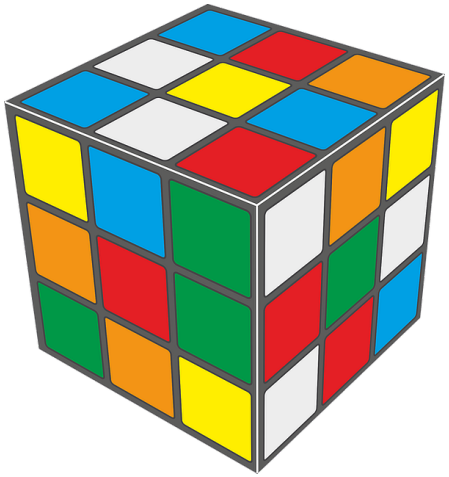
reinforcement learning example

Example



Q: *What would be a state/action/reward here?*

Example



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Objective model:

Reinforcement learning can be used to:

- select the optimal action when arriving at a state: $\underline{\mathbf{a}}^* = \underline{\mathbf{f}}(\underline{\mathbf{x}}, r)$
- find the transition model $P(\underline{\mathbf{x}}^{(t)} \,|\, \underline{\mathbf{x}}^{(t-1)}, \underline{\mathbf{a}}^{(t-1)})$, where t is a time step within the sequence.

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Q: *Would you regard reinforcement learning as supervised or unsupervised learning?*

	Data	Objective model
Supervised learning	$\left(\underline{\mathbf{x}}^{(1)}, \underline{\mathbf{y}}_T^{(1)}\right), \ldots, \left(\underline{\mathbf{x}}^{(p)}, \underline{\mathbf{y}}_T^{(p)}\right)$ ↳ label $\mathbb{R}$, $\mathbb{R}^M$,$\{0,1\}$, $\{0,\ldots,K-1\}$ ↳ observation $\mathbb{R}^N$, often iid.	discrimination function $\underline{\mathbf{y}} = \underline{\mathbf{f}}(\underline{\mathbf{x}})$ conditional distribution $P(\underline{\mathbf{y}}\|\underline{\mathbf{x}})$
Unsupervised learning	$\underline{\mathbf{x}}^{(1)}, \ldots, \underline{\mathbf{x}}^{(p)} \in \mathbb{R}^N$ observations often iid.	generative model/ data distribution $P(\underline{\mathbf{x}})$ clustering: $f(\underline{\mathbf{x}}) : \mathbb{R}^N \mapsto \{0,\ldots,K-1\}$ dimensionality reduction: $\underline{\mathbf{f}}(\underline{\mathbf{x}}) : \mathbb{R}^N \mapsto \mathbb{R}^M$
Reinforcement learning	$\left(\underline{\mathbf{x}}^{(0)}, \underline{\mathbf{a}}^{(0)}, r^{(0)}\right), \ldots, \left(\underline{\mathbf{x}}^{(p)}, \underline{\mathbf{a}}^{(p)}, r^{(p)}\right)$ ↳ reward $r \in \mathbb{R}$ ↳ action $\underline{\mathbf{a}} \in \mathbb{R}^M$ or $\{0,1\}^A$ ↳ state $\underline{\mathbf{x}} \in \mathbb{R}^N$ or $\{0,1\}^S$	select optimal action: $\underline{\mathbf{a}}^* = \underline{\mathbf{f}}(\underline{\mathbf{x}}, r)$ transition model: $P(\underline{\mathbf{x}}^{(t)} \mid \underline{\mathbf{x}}^{(t-1)}, \underline{\mathbf{a}}^{(t-1)})$

Setting expectations

Learning Paradigms

Connectionist Neuron (Perceptron)

References

Connectionist Neuron (Perceptron)

A neuron is a computational unit for processing information.

Q: *What does a connectionist neuron compute?*

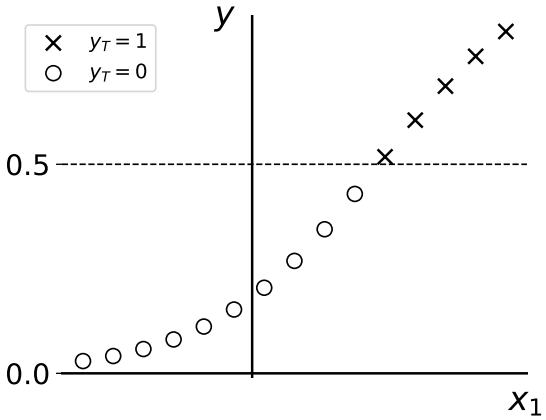
Connectionist Neuron (Perceptron)

A neuron is a computational unit for processing information.

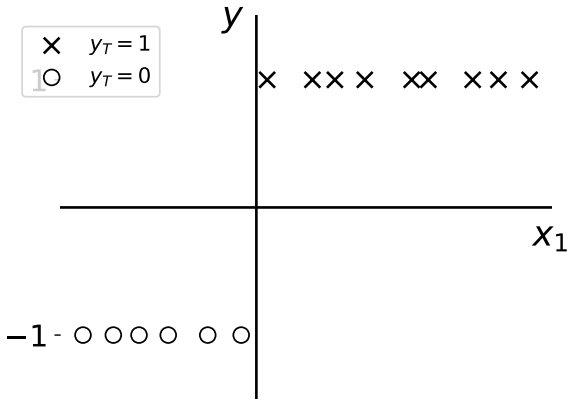
Q: *What does a connectionist neuron compute?*

- A connectionist neuron is a type of neuron model which measures for a feature within an observation (a data point). It is a feature detector/extractor e.g. edge detector, above or below a threshold.

A connectionist neuron's response to 1D data



(a)



(b)

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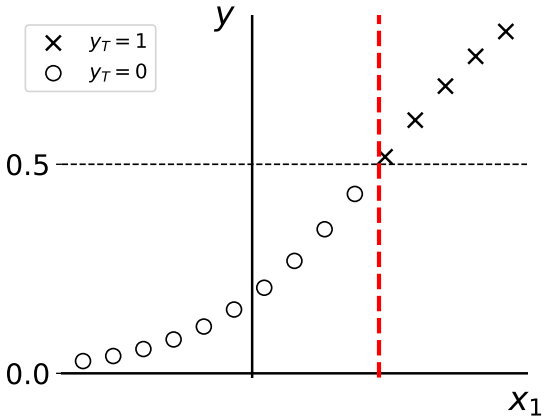

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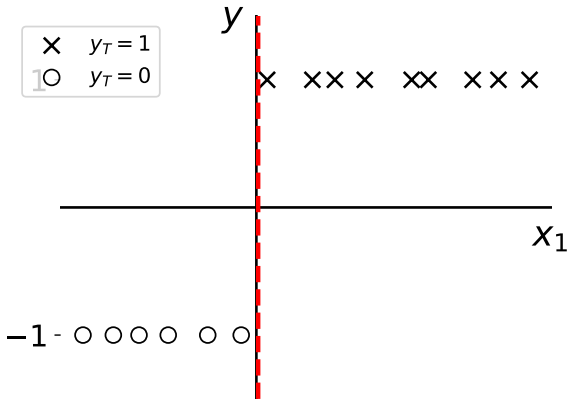

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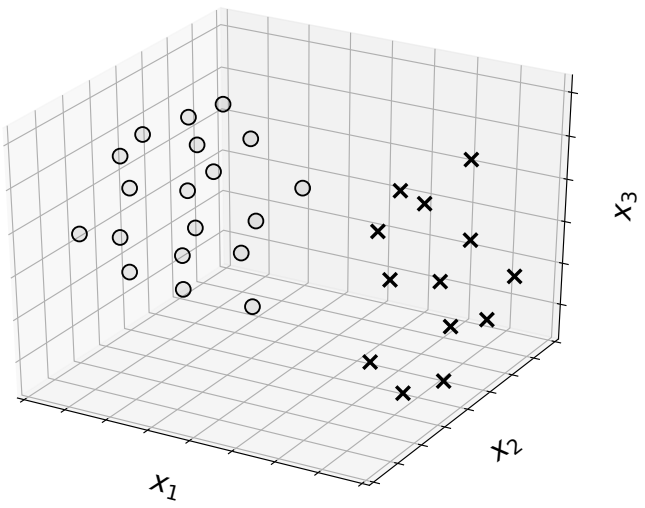
A connectionist neuron's response to 3D data:



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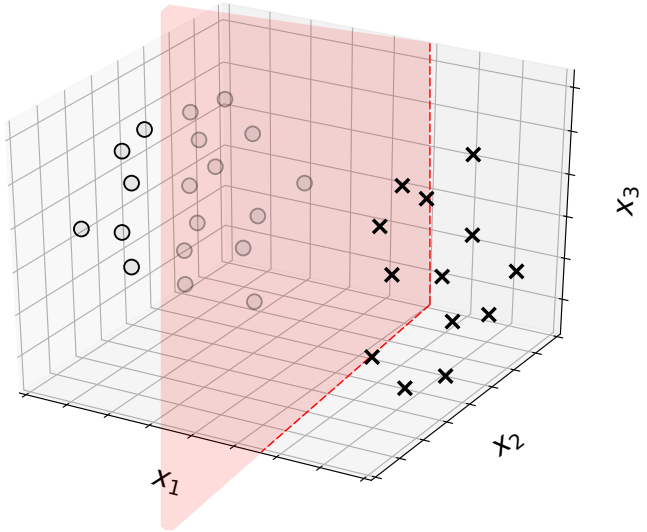


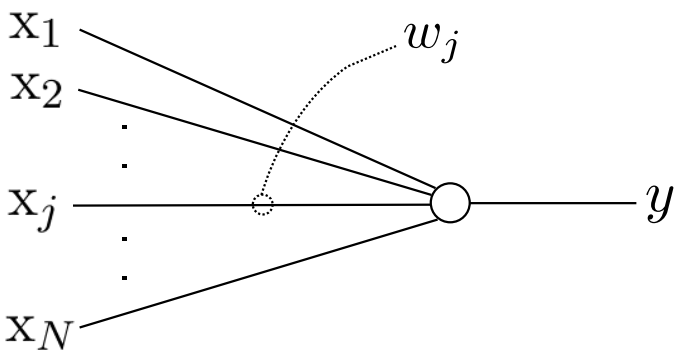
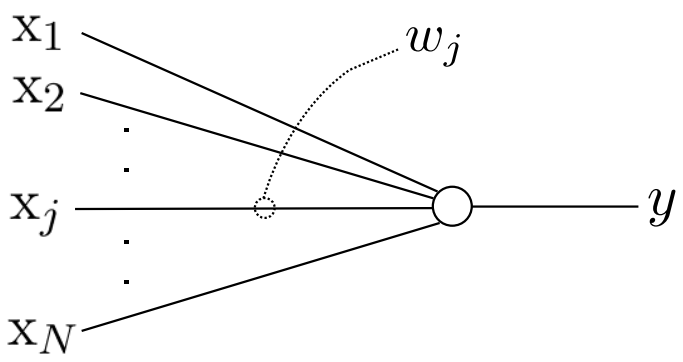
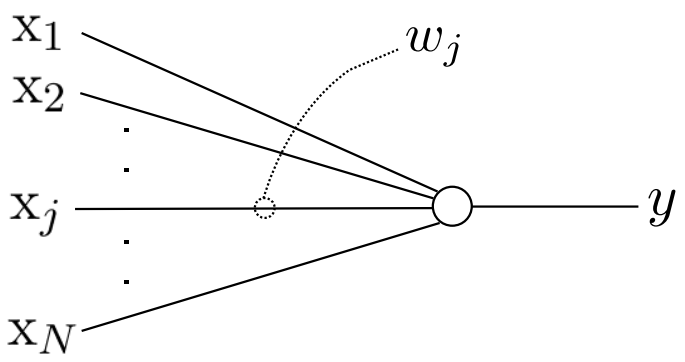


Figure: A graphical representation of the input-output relationship for a connectionist neuron. w_j describes the connection from the j-th input.



Given an input vector $\underline{\mathbf{x}} \in \mathbb{R}^N$ with components x_j, the connectionist neuron is composed of the following elements:

a. weights $\underline{\mathbf{w}} \in \mathbb{R}^N$
b. A linear filter: $\underline{\mathbf{w}}^\top \underline{\mathbf{x}} = \sum_j w_j x_j$.
c. A bias value $\theta \in \mathbb{R}$ also known as the *threshold* of the neuron.
d. An activation function or transfer function $f : \mathbb{R} \mapsto \mathbb{R}$.
e. The scalar output of the neuron: y.



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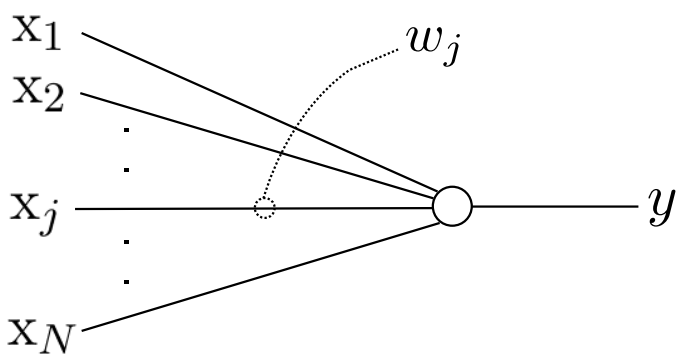
Q: *How does one interpret the scalar product* $\underline{\mathbf{w}}^\top \underline{\mathbf{x}}$*?*



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$$\underline{\mathbf{w}}^\top \underline{\mathbf{x}} = ||\underline{\mathbf{w}}||\ ||\underline{\mathbf{x}}||\ \cos\beta$$

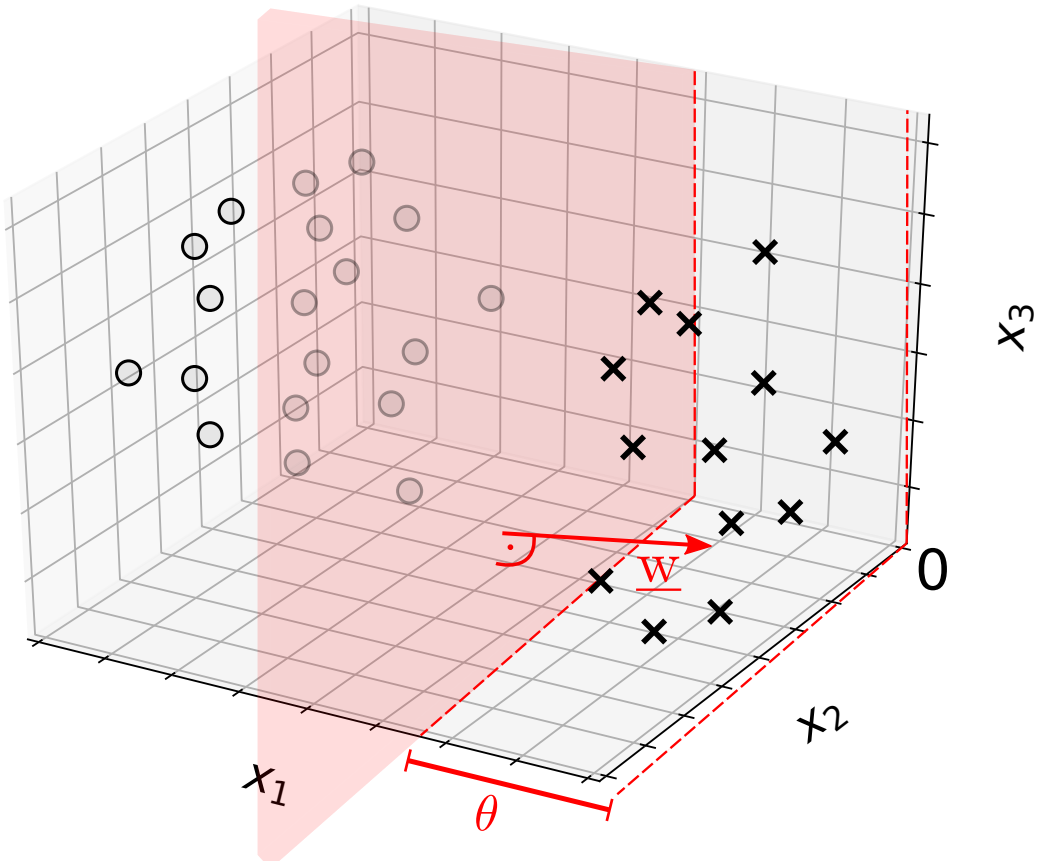


Figure: The plane divides the responses of the neuron. In the context of classification it is referred to as the decision boundary.

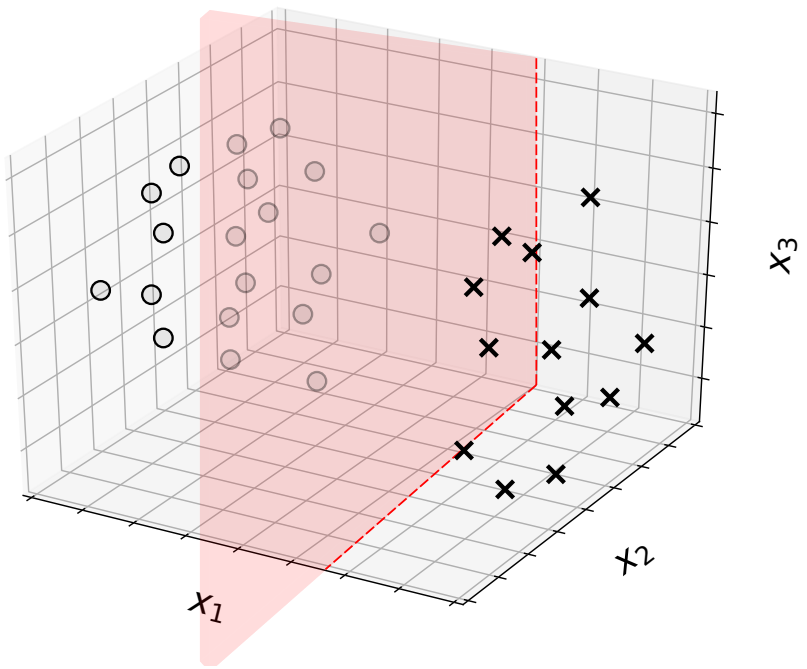
The input-output relationship

The input-output relationship is described by a linear filter with a static non-linearity $f(\cdot)$:

$$y = f\Big(\underbrace{\sum_{j=1}^{N} w_j x_j - \theta}_{=:h} \Big) = f\big(\underline{\mathbf{w}}^\top \underline{\mathbf{x}} - \theta \big) = f(\, h \,)$$

$\underline{\mathbf{x}}$:	input vector with components x_j
y :	scalar output of the neuron
$\underline{\mathbf{w}}$:	weight vector of the neuron with components w_j
θ :	threshold of the neuron
h :	total input of the neuron.
	$h \propto \frac{\underline{\mathbf{w}}^\top \underline{\mathbf{x}}}{\|\|\underline{\mathbf{w}}\|\|_2}$ component of $\underline{\mathbf{x}}$ in the direction of $\underline{\mathbf{w}}$
$f(\cdot)$:	transfer function

Q: *What roles do the weights and bias play?*



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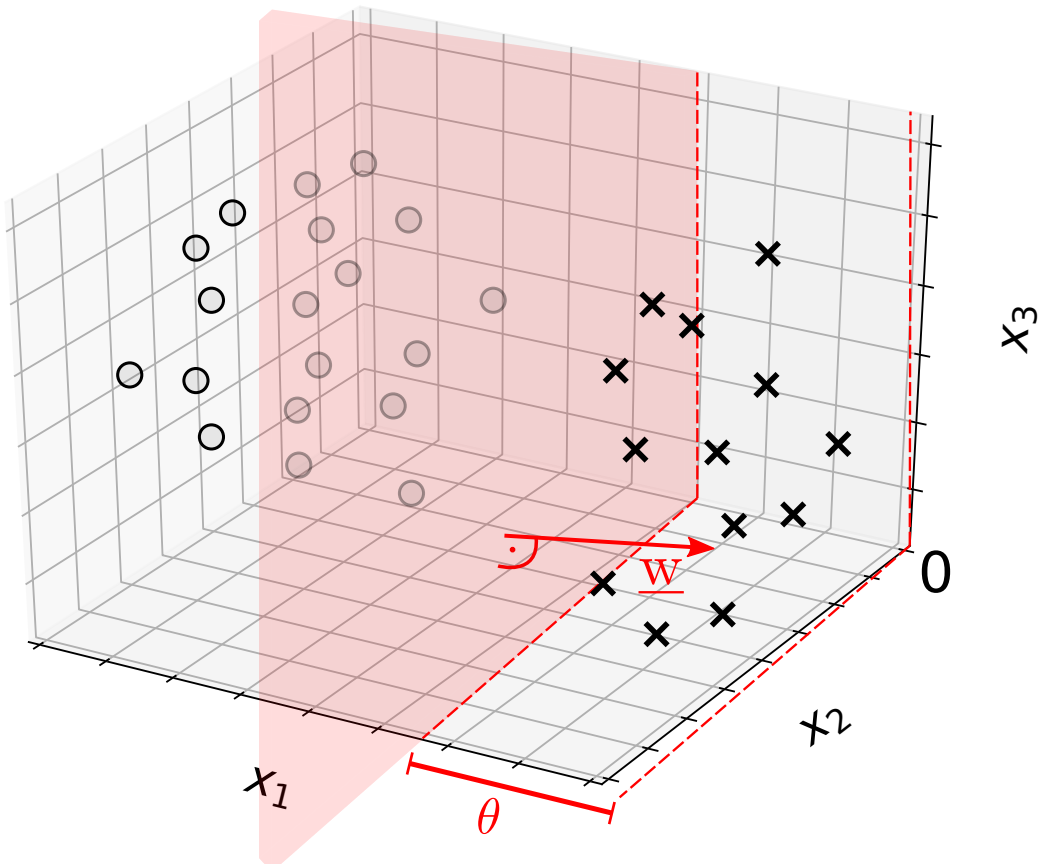
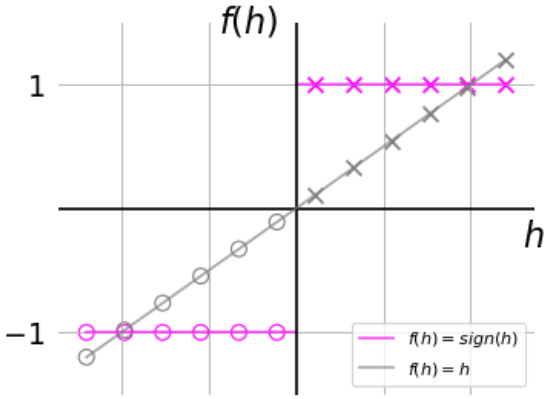
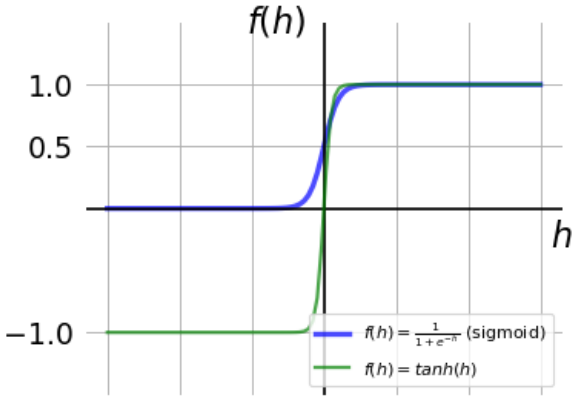


Transfer functions

Q: *What are the advantages of using a non-linear transfer function instead of a linear one?*



(a) Linear vs. non-linear activation



(b) logistic sigmoidal $f(h)=\frac{1}{1+e^{-h}}$ vs. tanh

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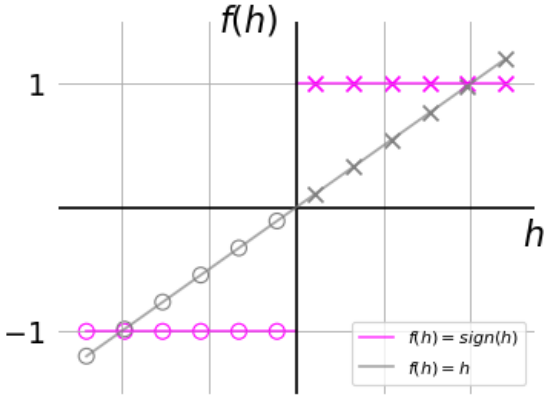

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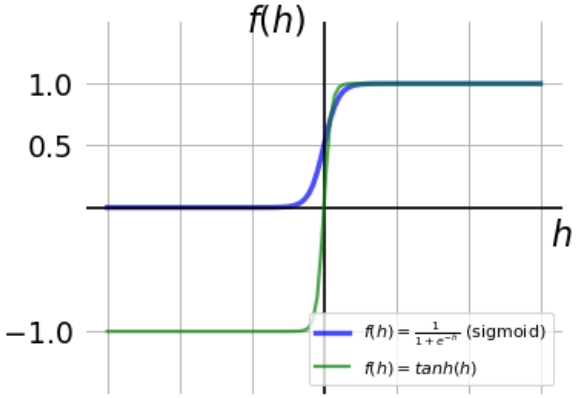

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- Advantages:

1. binary classification, either $f(h) \in \{0, 1\}$ or $f(h) \in (-1, 1)$

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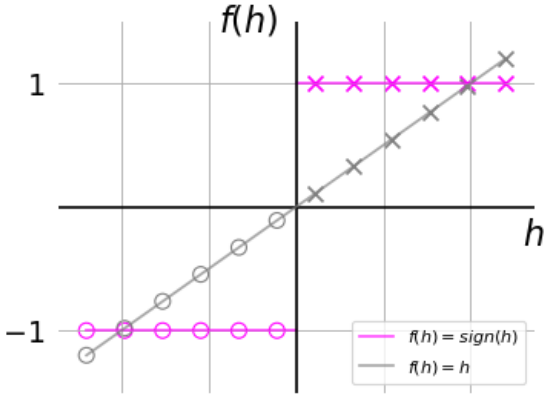

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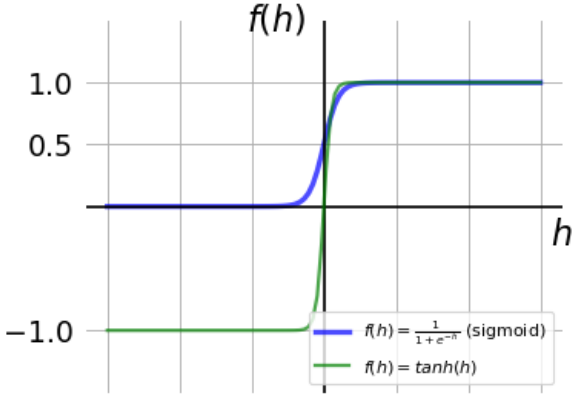

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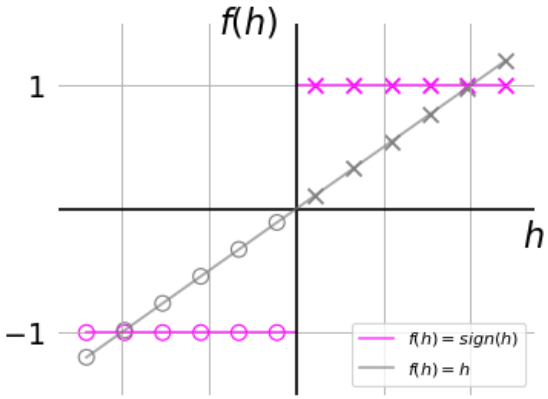
- Advantages:

2. interpret $f(h)$ as a probability. The logistic sigmoidal where $f(h) = \frac{1}{1+exp(-h)}$ yields values in the range of (0,1).

Transfer functions

Q: *What are the advantages of using a non-linear transfer function instead of a linear one?*



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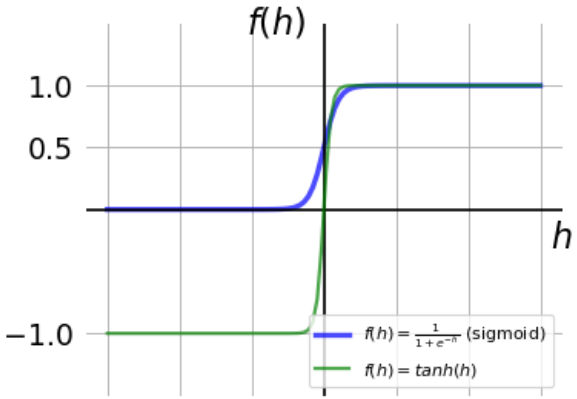

(b) logistic sigmoidal $f(h)=\frac{1}{1+e^{-h}}$ vs. tanh

- Advantages:

3. a multilayer perceptron with only linear transfer functions can be reduced to a single layer.[1]:

[1] Recent literature reveals interesting insights into what kind of representations arise during learning across the layers of linear networks. For more information see (**?**).

Shortcut notation for weights and bias

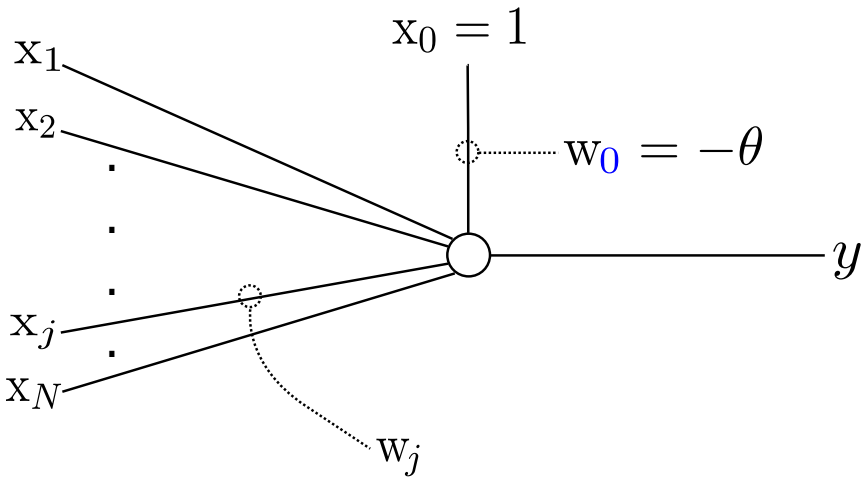

Figure: Absorb bias into weight vector.

$$y = f\Big(\sum_{j=0}^{N} w_j x_j\Big) = f(\underline{\mathbf{w}}^\top \underline{\mathbf{x}})$$

Shortcut notation for weights and bias

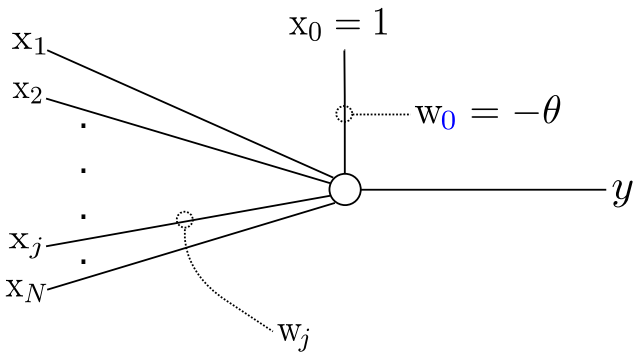

Figure: Absorb bias into weight vector.

$\underline{\mathbf{w}}$ will be used for $\begin{pmatrix} w_1 \\ \vdots \\ w_N \end{pmatrix}$ as well as for $\begin{pmatrix} w_0 \\ w_1 \\ \vdots \\ w_N \end{pmatrix}$.

Accordingly, $\underline{\mathbf{x}}$ will be used for $\begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix}$ as well as for $\begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_N \end{pmatrix}$.

References I

Andrew M Saxe, James L McClelland, and Surya Ganguli. A mathematical theory of semantic development in deep neural networks. *Proceedings of the National Academy of Sciences*, 116 (23):11537–11546, 2019.